Qatar-2b

R-0780

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_240214` · created 2026-07-28T06:06:09+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460354.9881 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.027
Transit depth (Rp/Rs)^2	3.25 +/- 0.96 [%]
Orbital Inclination (inc)	87.19 +/- 1.61
Airmass coefficient 1 (a1)	1.071 +/- 0.021
Airmass coefficient 2 (a2)	-0.048 +/- 0.081
Scatter in the residuals of the lightcurve fit is	1.92 %
Best Comparison Star	#3 - [199, 243]
Optimal Method	PSF photometry
Transit Duration (day)	0.0745 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.027 vs 0.16327 (deviation 0.54σ)
Duration fit vs published	0.0745 d vs 0.0767 d (deviation 0.47σ)
Mid-time O–C	3.9 min
Depth SNR	5.8
Residual scatter	1.92 %

Light curve

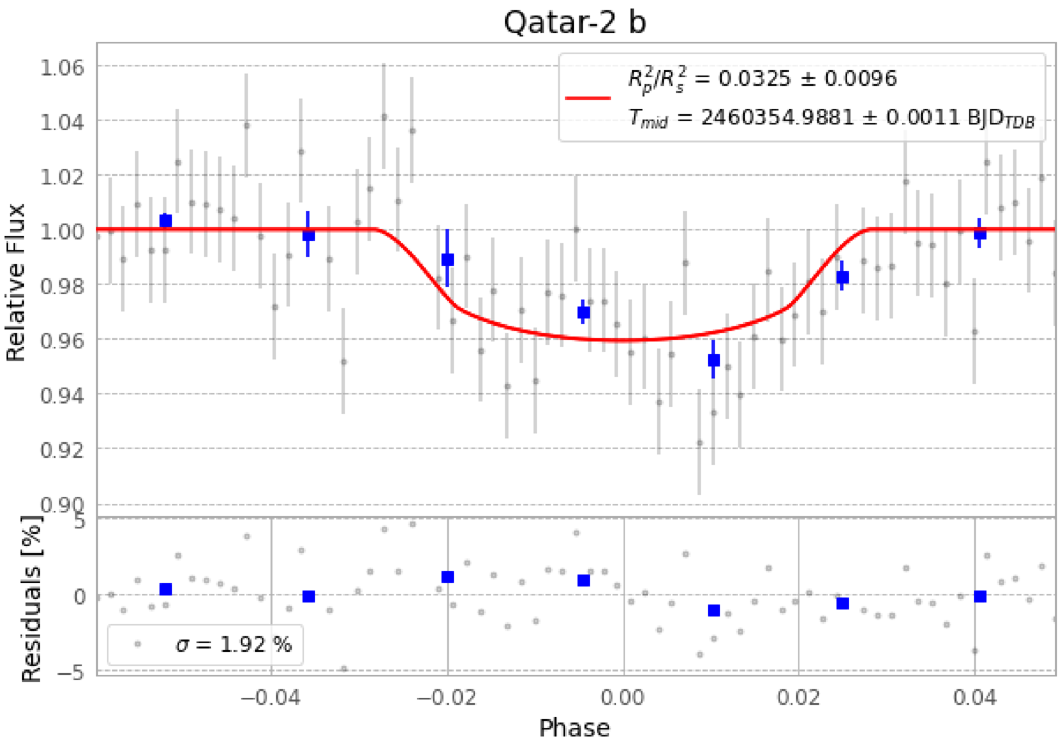

Figure 1 — FinalLightCurve_Qatar-2 b_2024-02-14.png (SHA-256 bb7a810ced79b586…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [549, 181], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 eb4db8af3d1ea796…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

71 FITS frames (47 MB), `Qatar-2240214094215.FITS` → `Qatar-2240214131215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240214.log (cb37e68e6aff…), FinalLightCurve_Qatar-2 b_2024-02-14.png (bb7a810ced79…), FinalLightCurve_Qatar-2 b_2024-02-14.csv (44cf2ebf5ead…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)